

GIS Project 1

2107ENV, Trimester 1, 2021

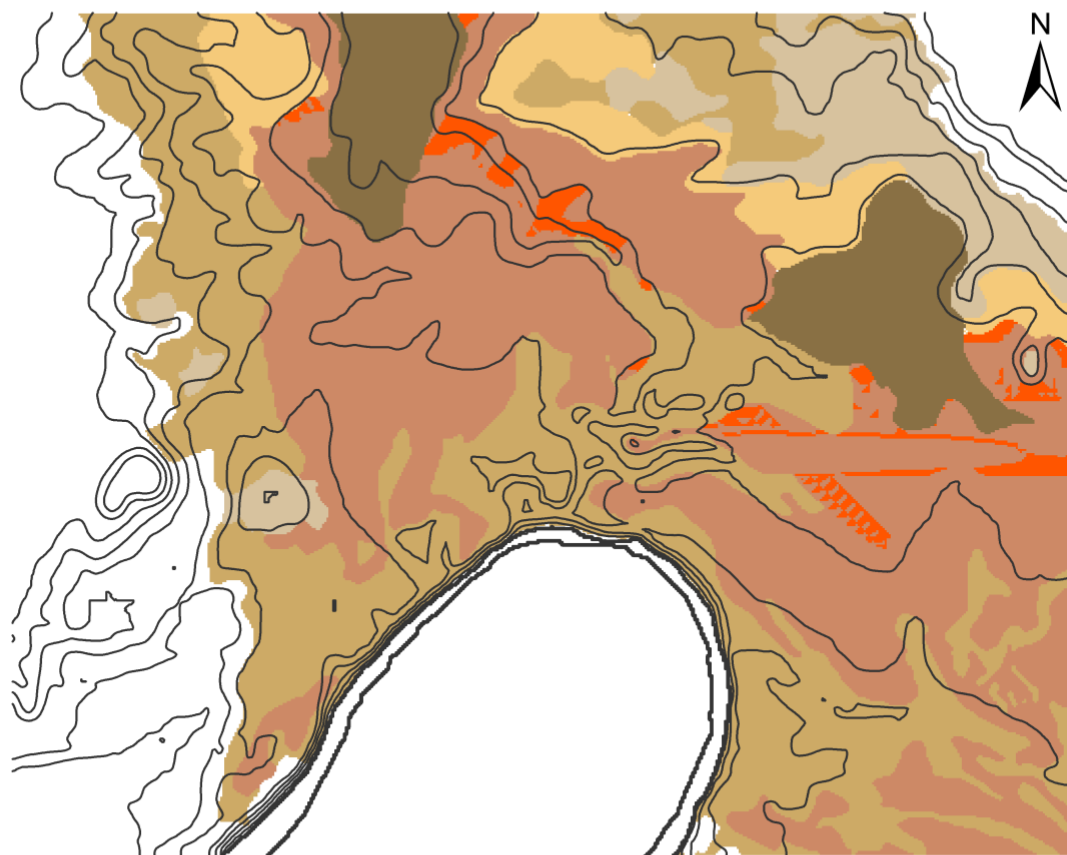
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Area Suitable for Recessional Agriculture 8 Metre Flood Zone



0 0.5 1 2 3 4
Kilometers

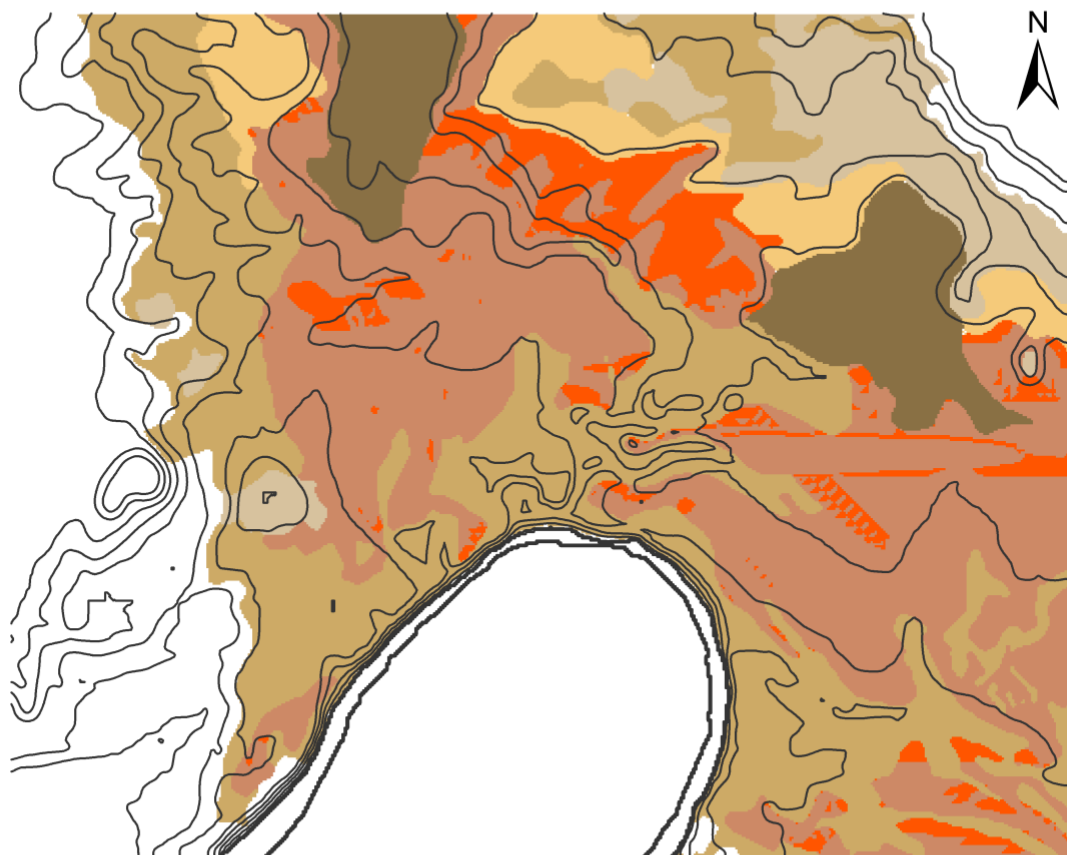
Legend

— contour

- 1
- 2
- 3
- 4
- 5
- 6

Figure 1: Map Displaying Area Suitable for Agriculture in an 8 Metre Flood Zone

Area Suitable for Recessional Agriculture 10 Metre Flood Zone



0 0.5 1 2 3 4 Kilometers

Legend

— contour

- 1
- 2
- 3
- 4
- 5
- 6

Figure 2: Map Displaying Area Suitable for Agriculture in a 10 Metre Flood Zone

Discussion:

Analysis of the two developed maps displaying the areas suitable for recessionary agriculture in 8 metre and 10 metre flood zones – denoted by **Class 6** (Refer to the Legend), revealed the following findings. Firstly, in both maps, the areas suitable for recessionary agriculture appear to mirror the appearance of the contour lines, which depict and describe the slope and aspect of the region. This is feasible as the provided criteria outlined that the suitable area must be evident within regions possessing a slope of less than 3 degrees as well as being on a Southern facing aspect. Furthermore, in order to adhere to the given conditions, the suitable agriculture areas evident on both established maps would contain normal clay soil. Ultimately, comparison of the two established maps indicates that when employing a flood threshold of 10 metres, a noticeable increase in area suitable for recessionary agriculture is generated. Thus, in order to maximise land suitable for use in agriculture, it is recommended a 10-metre flood threshold is used.

Employing the attribute table of the developed layer “*bestarea3*” to obtain the pixel count of each suitable area as well as the fact that a unit pixel is 90 m^2 , the size of each area (in square kilometres) suitable for recessionary agriculture was determined.

8 Metre Flood Zone:

$$\text{Area} = 3151 \times (30 * 30)$$

$$\text{Area} = 2835900\text{ m}^2$$

$$\text{Area} = 2.8359\text{ km}^2$$

10 Metre Flood Zone:

$$\text{Area} = 9588 \times (30 * 30)$$

$$\text{Area} = 8629200\text{ m}^2$$

$$\text{Area} = 8.6292\text{ km}^2$$

As displayed by the calculations presented above, when the flood threshold is increased from 8 metres to 10 metres, the area suitable for recessionary agriculture increases by a factor of 3.04.